



CBG-CGD SYNCHRONISATION

KEC CBG -CGD STUDY

Kec Agritech



CBG-CGD SYNCHRONISATION



1. What is the CBG-CGD Synchronisation Scheme?

The CBG-CGD Synchronisation Scheme (Synchro Scheme) is a scheme launched by the Ministry of Petroleum & Natural Gas (MOP&NG) for synchronisation of Biogas/Compressed Biogas (CBG) in CGD networks. Under the Scheme, the Biogas/CBG is required to meet applicable technical/quality standards i.r.o CGD networks for supply to CGD entities in nearby Geographical Areas (GAs). GAIL has been mandated to operationalize the CBG-CGD Synchronisation Scheme and supply Biogas/CBG co-mingled with domestic gas at Uniform Base Price (UBP) to all CGD entities for use in CNG (T) and PNG (D) segments

2. Can Biogas/CBG supply under Synchro Scheme be made only through injection in the CGD pipeline network?

No, under the Synchro Scheme both Biogas and Compressed Biogas (CBG) can be supplied. Compressed Biogas (i.e. CBG) @pressure of 200 – 250 bars may supply through Cascades for sale through Retail Outlets. Biogas may also be injected in the pipeline network of concerned CGD entities @the pipeline pressure (~7 Bar for MDPE network and between 16 to 49 Bar for Steel network).

3. Do I have to get an LoI from any Oil & Gas Marketing Company (OGMC) to supply my Biogas/CBG under the Synchro Scheme?

No. There is no need to have an LoI from any OGMC to supply Biogas/CBG under the Synchro Scheme. A CBG producer just needs to sign an agreement with GAIL to sell his Biogas/CBG to GAIL and also sign a Tripartite Agreement (TPA) between CBG Producer, GAIL and nearby CGD entity to enter into the Synchro Scheme.

4. How can I supply my Biogas/CBG to the nearby CGD entity under the Synchro Scheme?

To enter into the Synchro Scheme, the nearest Zonal Office of GAIL may be contacted (contact details of GAIL Zonal Offices), discuss the contractual modalities with GAIL and the concerned CGD entity, sign the requisite contracts and begin supplying your Biogas/CBG under the Synchro Scheme.

5. How is the Biogas/CBG quantity accounted under the Synchro Scheme?

GAIL has taken care to build operational excellence in various aspects of the Synchro Scheme while implementing the Scheme. For accounting the Biogas/CBG purchased and sold, a web-portal for online joint ticketing has been developed and is available to CBG producers and CGD entities. This portal can be accessed securely and in a transparent manner by the CBG producers and CGD entities through their login credentials for entering the daily quantity supplied and accepted.

6. How is the Biogas/CBG quantity invoiced under the Synchro Scheme?

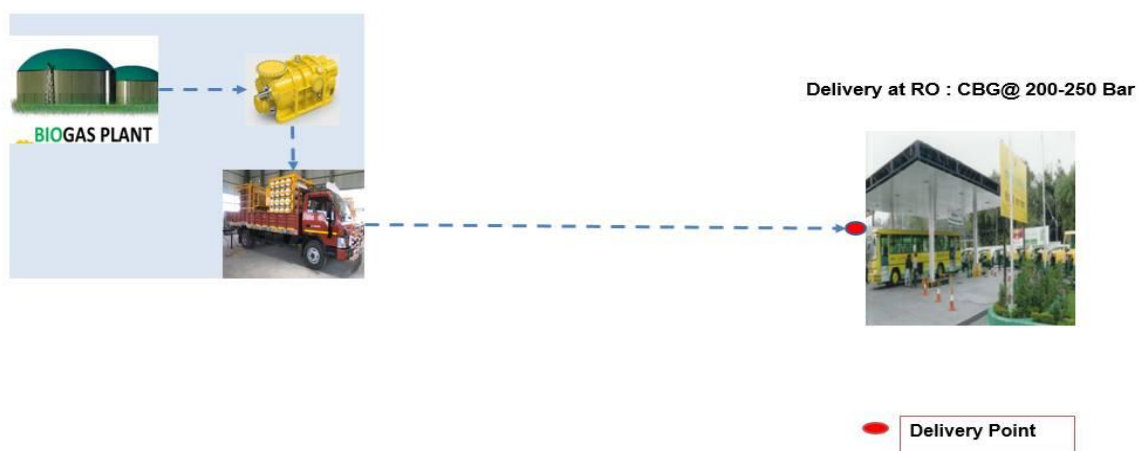
Based on the fortnightly quantity of Biogas/CBG supplied by the producer and accepted by CGD entity, as captured in the online joint-ticketing web portal, the CBG producer raises a fortnightly invoice on GAIL. Further, a fortnightly invoice is also raised upon the CGD entity by GAIL based on joint-ticket for realizing payments for Biogas/CBG supplied.

Quality Specifications of Biogas (Bio-Methane) IS 16087:2016

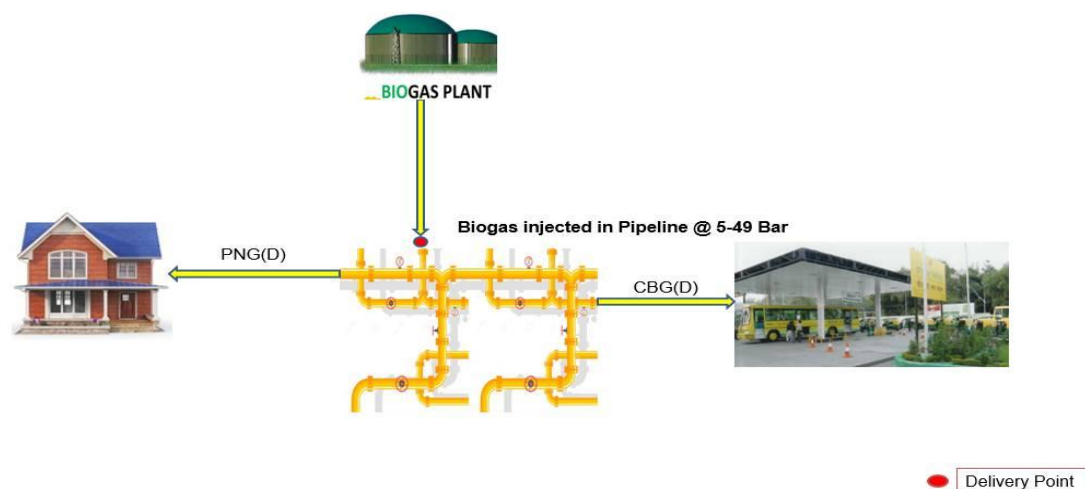
Sl. No.	Characteristic	Requirement
i	Moisture, mg/m ³ (Max.)	5
ii	Methane, % (Min.)	90.0
iii	Total Sulphur (including H ₂ S), mg/m ³ (Max.)	20
iv	CO ₂ + N ₂ + O ₂ , % (v/v) (Max.)	10
v	Only CO ₂ , % (v/v) (Max.)	4
vi	Oxygen, % (Max.)	0.5

Delivery Options

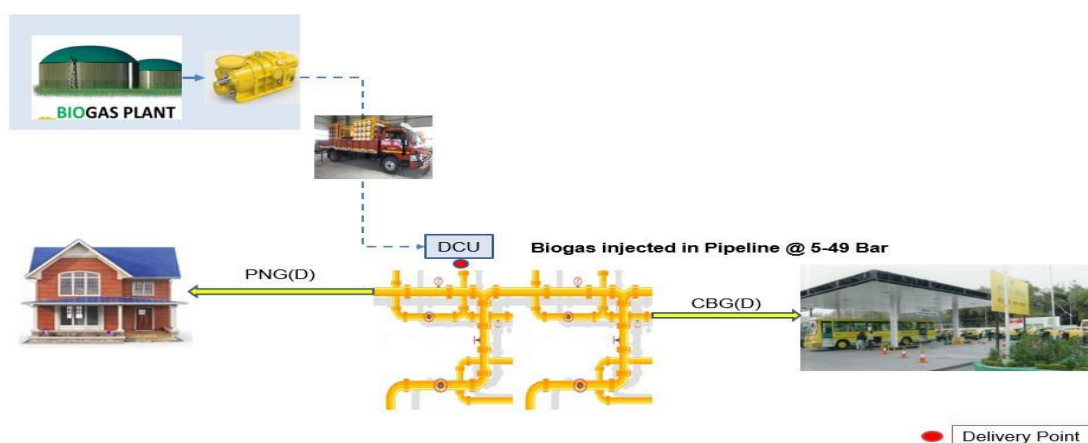
1. CBG Delivery @ 200 – 250 Bar pressure in Cascade for delivery at Retail Outlets



2. Biogas Injected in CGD Pipelines Network @ Pipeline Pressure



3. Biogas Injected Through DCU in CGD Pipelines Network @ Pipelines Pressure



Report Disclaimer

The specifications, policies, and commercial terms referenced in this report are based on information available from official GAIL and Ministry of Petroleum & Natural Gas (MoPNG) sources as on the date of preparation. Stakeholders are advised to verify the latest notifications, BIS standards, PNGRB regulations, SATAT guidelines, and GAIL circulars before undertaking project development, financing, engineering design, commercial negotiations, or execution activities, as regulatory and technical requirements may be revised from time to time.

Note: - The proposed CBG project shall be designed to produce biomethane conforming to the quality standards prescribed by BIS IS 16087 and GAIL requirements. The produced CBG may be supplied through cascades, retail outlets, or injected into CGD networks subject to compliance with applicable technical, safety, and contractual requirements under the CBG-CGD Synchronization Scheme.